

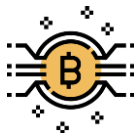


Blockchain and Distributed Ledger in 2021 continued to attract the interest of institutions, companies and media around the world. The market continues to evolve in different directions.

Applications of Blockchain technology

Internet of Value

Applications focused on value exchange, such as cryptocurrencies, stablecoins, and virtual currencies promoted by central banks - CBDCs



Blockchain for business

Projects where traditional business processes are replicated using Blockchain technologies



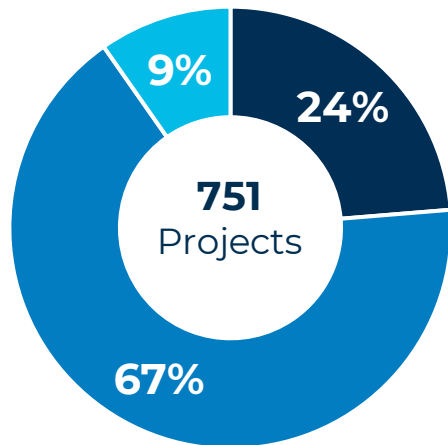
Decentralized web

Blockchain becomes an enabling infrastructure for the creation and development of innovative business solutions

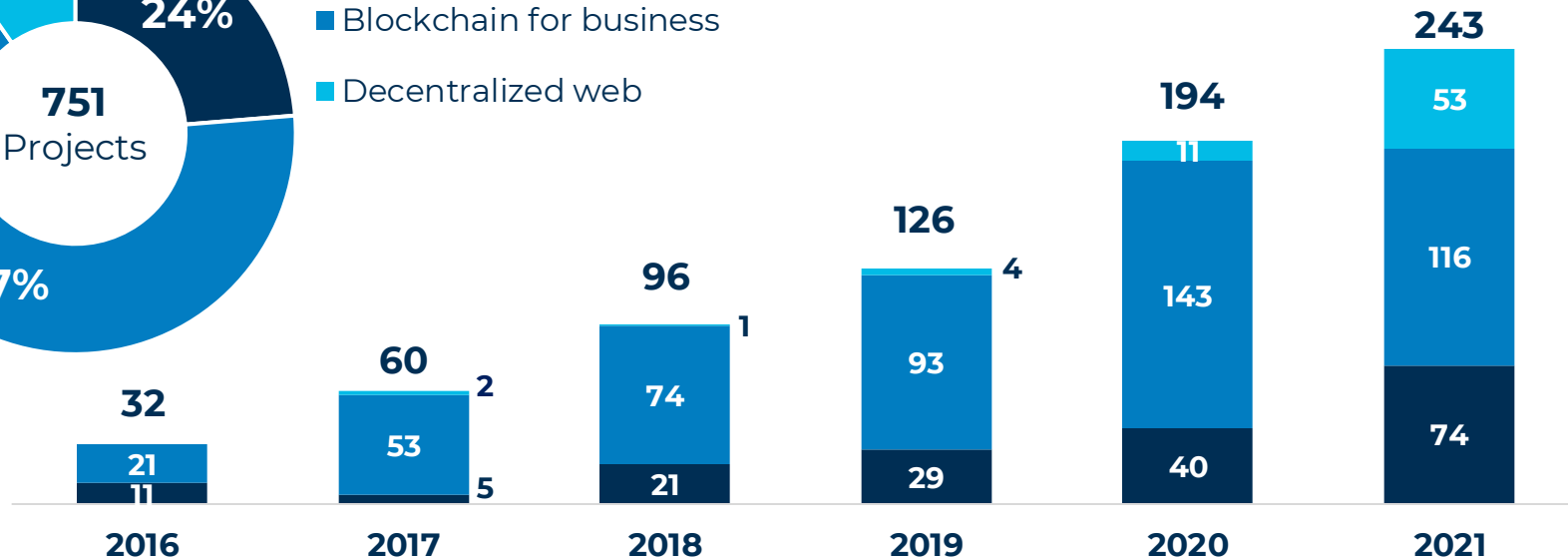




Blockchain and Distributed Ledger in 2021 continued to attract the interest of institutions, companies and media around the world. The market continues to evolve in different directions.

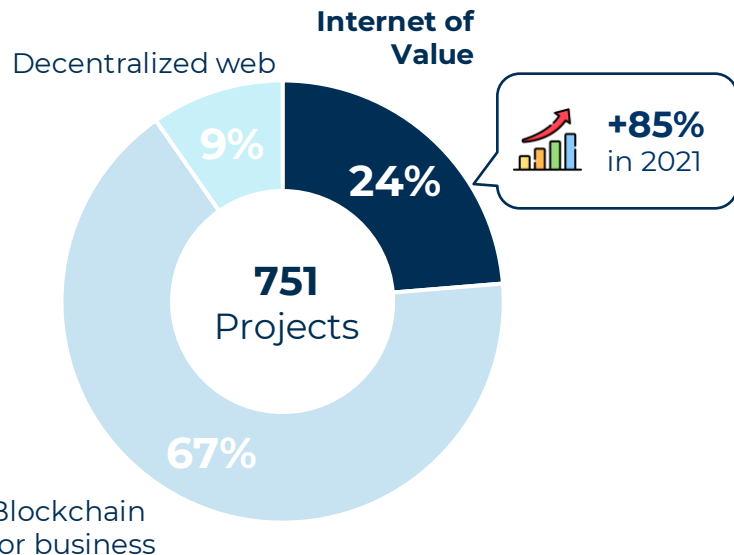


- Internet of Value
- Blockchain for business
- Decentralized web





2021 has been a maturing year for cryptocurrencies. The attention of institutions and some large companies has also focused on stablecoin and CBDC, pending regulatory evolution (MICAr).



Cryptocurrency



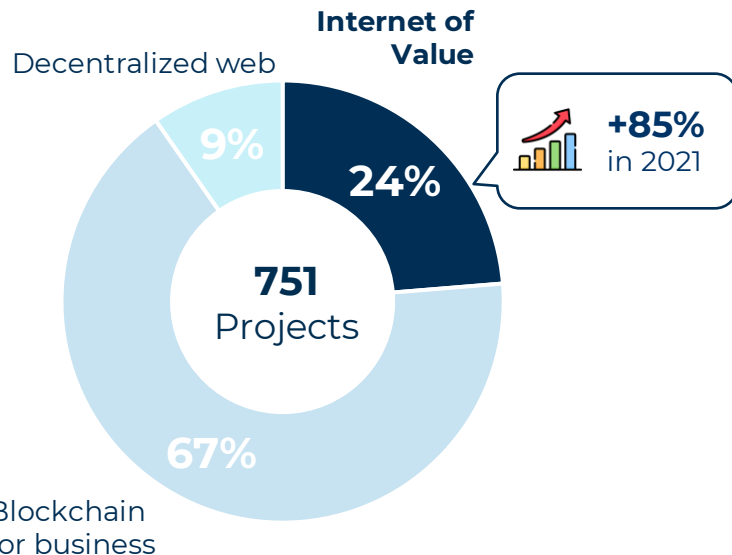
Stablecoin



CBDC



2021 has been a maturing year for cryptocurrencies. The attention of institutions and some large companies has also focused on stablecoin and CBDC, pending regulatory evolution (MICAr).



- Coinbase listing on Nasdaq
- Partnerships between crypto startups and institutional players increase
- Traditional actors begin to integrate cryptocurrencies
- Crypto actors invest in big marketing sponsorships

coinbase





?

**What is the state of adoption of
cryptocurrencies, stablecoins and CBDCs
by traditional players?**



We want to investigate 4 different types of players who have taken steps towards the Internet of Value.

Product
and
service



Banks



Payment
providers

Users



Private
users



Merchants



We want to investigate 4 different types of players who have taken steps towards the Internet of Value.

Analysis of the projects of the **top 100 banks** in the world



Banks



Payment providers

Survey on the **Italian population** and **data from ECB**



Private users

Different methodologies



Merchants

Scouting of all the international **news related to IoV projects** from secondary sources



We want to investigate 4 different types of players who have taken steps towards the Internet of Value.



Banks



Payment
providers



Private
users



Merchants



BORDERS

- World



CRITERIA

- Top 100 largest banks for total assets under management
- Data at 11/04/2022



SOURCE

- S&P Global
- News on IoV projects from websites both general (e.g. Reuters, FT...) and specialized (e.g. Coindesk, Cointelegraph...)



100
World's largest
banks

Internet of Value: the banks' projects



CBDC

Projects related to the issuance, use, custody of Central Bank Digital Currencies



Stablecoin

Projects related to the use and adoption of stablecoins



Crypto

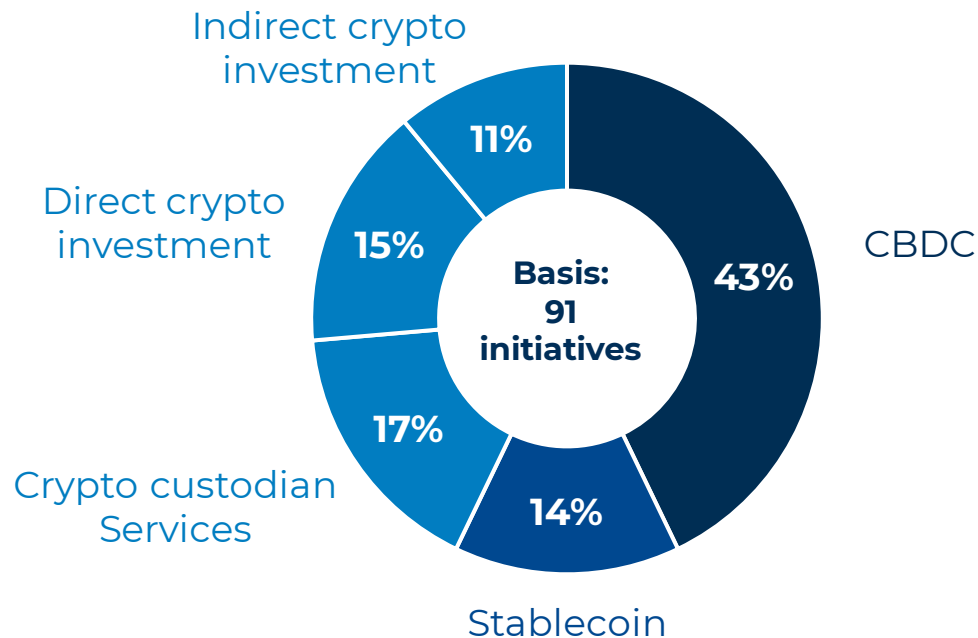
- **Direct crypto Investment:** the bank permits customers to buy cryptocurrencies
- **Indirect crypto investment:** the bank offers indirect exposure to cryptocurrencies through crypto funds or derivatives
- **Crypto custodian service:** the bank offers cryptocurrency custody services or develops a crypto wallet



59 banks
developed at least
one IoV project



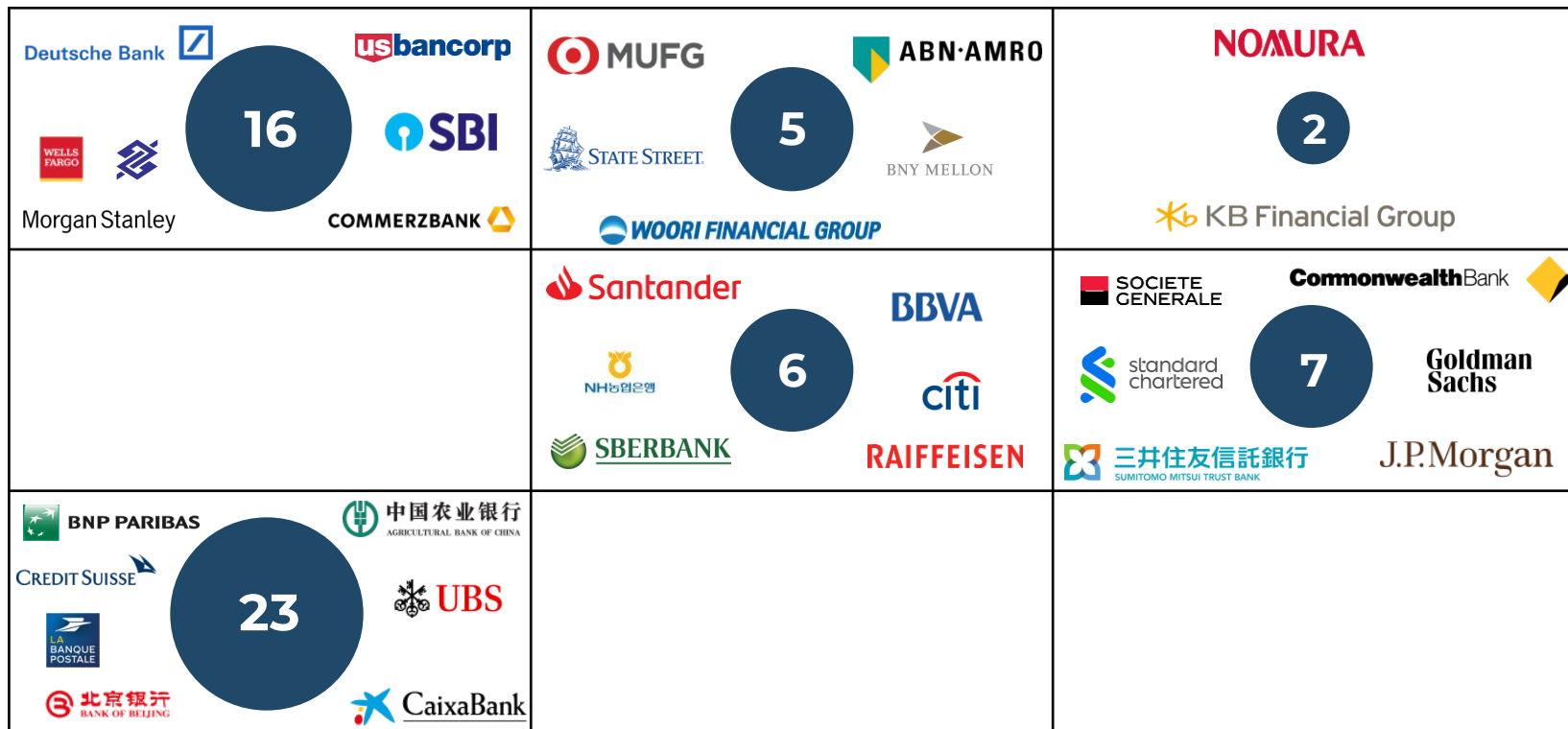
Most of the world's top banks are focused on **CBDC projects**



**Only
Crypto**

**CBDC
&
Crypto**

**Only
CBDC**



1 Project

2 projects

3+ Projects

20 Chinese banks in the sample are **prevented from trading cryptocurrencies** by the Central Bank of China

Additional **11 banks** have independently declared a **closure** toward **cryptocurrency operations**.



- Too high **risk** for customers
- **Regulation** uncertainty
- **Fraud** concerns

Unicredit, lo stop al correntista sui Bitcoin diventa un caso. Gli utenti: "Farete la fine di Blockbuster". La banca precisa: "Transazioni non inibite"

NatWest caps transfers to crypto exchanges, including Binance, over fraud concerns

Bank of America Has No Plans to Offer Crypto Services

BMO blocks cryptocurrency purchases on its credit, debit cards

Lloyds Bank bans customers from buying bitcoins using credit cards



We want to investigate 4 different types of players who have taken steps towards the Internet of Value.



Banks



Payment
providers



Private
users



Merchants



Many different players

Acquirers

Gateways

Card
schemes

Aggregators

Processors



Traditional payment providers are slowly adding **crypto**
and **stablecoins** to their **payment methods**.



Many different players

Acquirers

Gateways

Card
schemes

Aggregators

Processors



Some of them are also starting to introduce
custody services and access to **crypto investments**.





We want to investigate 4 different types of players who have taken steps towards the Internet of Value.



Banks



Payment
providers



Private
users

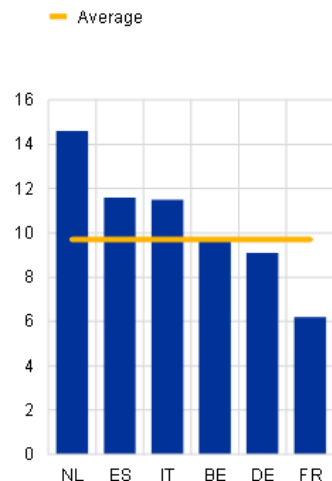


Merchants

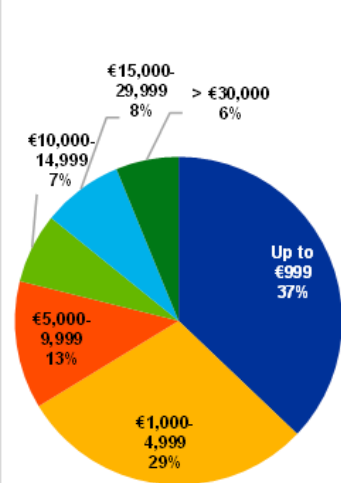


a) Share of respondents who reported that they or anyone in their household own crypto-assets

(percentages)

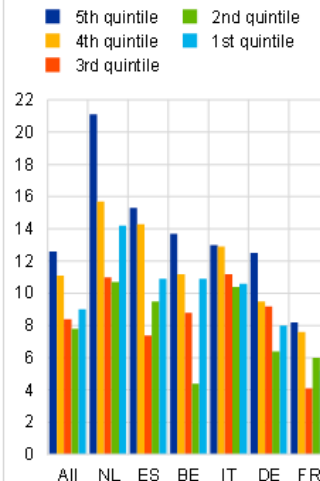


b) Crypto-asset owners' estimated holding values



c) Crypto-asset owners by income quintile

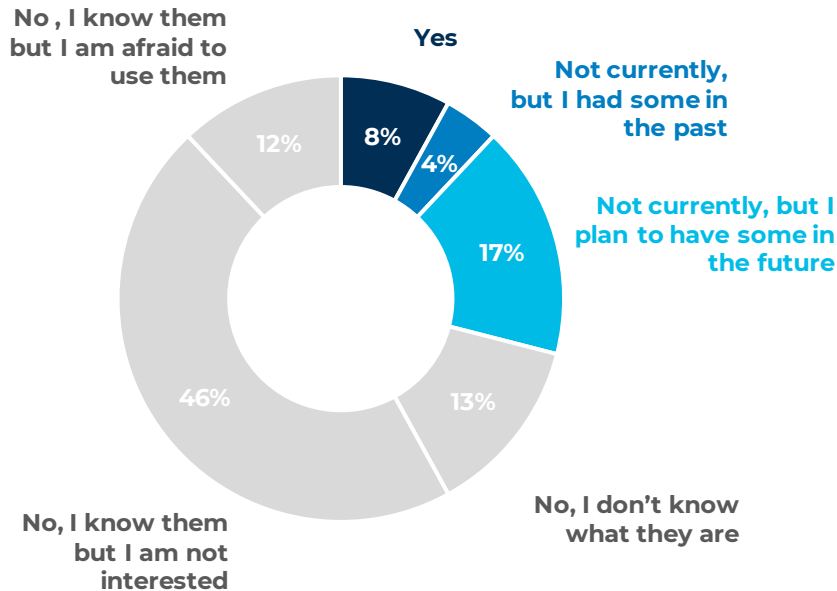
(percentages)



- a) As many as **10%** of households may own crypto-assets.
- b) Most crypto-asset owners reported holding **less than €5,000** in crypto-assets.
- c) Looking **at the income quintiles** of the respondents, **the pattern is largely U-shaped**

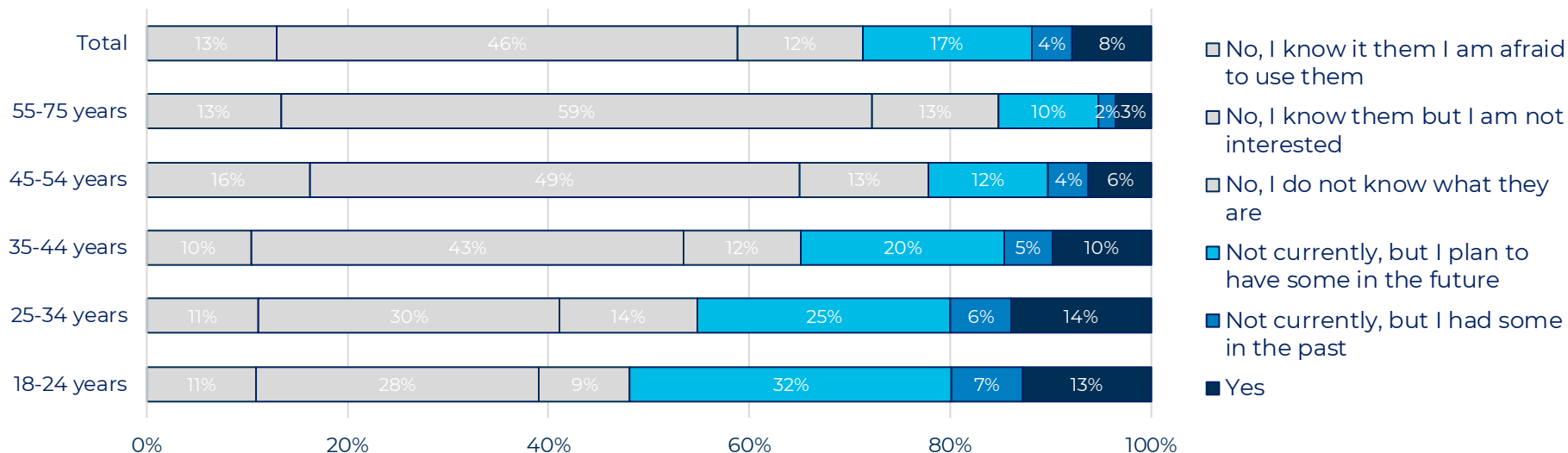
On average, **young adult males** and **highly educated** respondents were **more likely to invest** in crypto-assets in the countries surveyed.

«Do you own cryptocurrencies or tokens?»

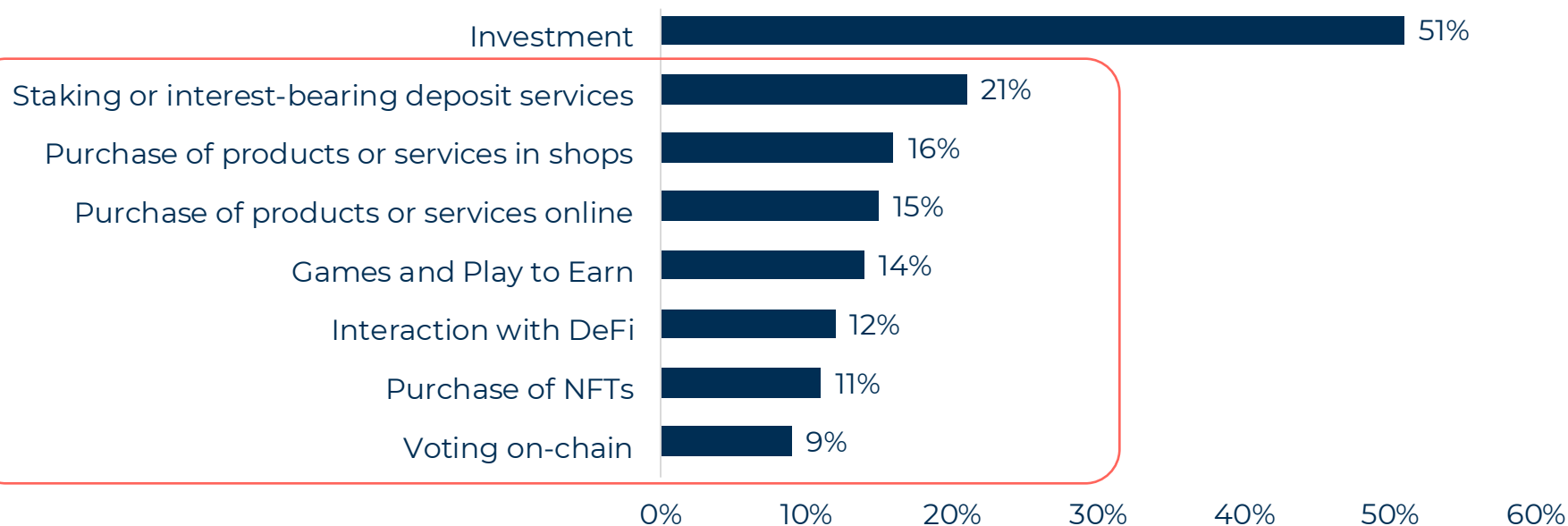


- **12% of people in Italy own or have owned** cryptocurrencies or tokens in the past.
- This percentage **could grow in the coming years**, considering that 17% of respondents say they are interested in buying them in the future.
- **67%** of who owns cryptocurrency or tokens is a **men**, only **33%** is a **woman**.

Almost half of the 18-24 years old population is projected to have crypto-assets in the future (*actual holders + future holders*).



Half of the respondents declared that they **use crypto-assets** for purposes **other than simple investment**





We want to investigate 4 different types of players who have taken steps towards the Internet of Value.



Banks



Payment
providers



Private
users

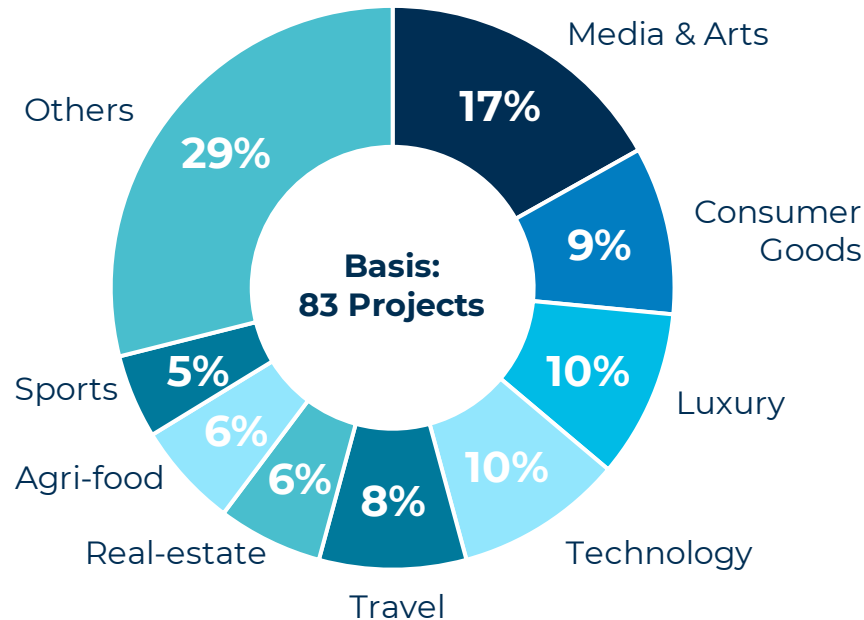


Merchants

Some big companies are starting to **accept cryptocurrencies and stablecoins** as a **payment method**.



Big companies accepting crypto and stablecoin



- The **main goal** for them is to **attract new customers**
- Most of these players **never directly receives cryptocurrencies**

The number of companies accepting crypto and stablecoin is **still very low**

On coinmap.org, e.g. only 29.760 crypto merchant are registered (+23% from last year).

In Italy merchants accepting crypto are **1.419**, over a total number of **1,7 million**.



Banks

Still few **banks are working on IoV projects.**

Most of those that are doing projects **focus on CBDC** and are not directly active on cryptocurrencies and stablecoins.



Payment providers

Traditional **payment providers adding crypto and stablecoins** to their **payment methods**. However their solutions are still limited and restricted to a small set of crypto.



Private Users

Users show a high interest in adopting crypto-assets with the share of crypto users set to grow in the future. Adoption may **not only be limited to investment** purposes.



Merchants

The **acceptance of crypto payments is still uncommon** and limited to a few large players. However, interest is high especially for attracting new customers.



Crypto payments provider



Challenger and Crypto banks



Digital asset custodian



Digital asset exchange



DeFi





Stablecoins are on-chain digital assets designed to maintain a stable value over time **against a target asset** (e.g., a fiat currency, a commodity, a financial instrument)



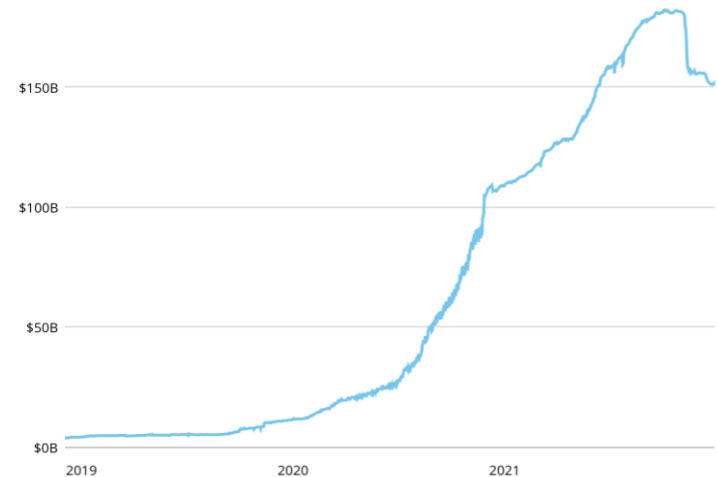
Stablecoins aim to be an efficient digital means of payment, providing the **upsides of cryptocurrencies** while **limiting the price volatility**

Stablecoins account for **17%** of the total cryptocurrencies **market cap**

Stablecoin transaction volumes for each quarter in 2021 were **\$1 billion**

Source: McKinsey.com

Stablecoin market cap has grown **+37%** in the **past 12 months**



Stablecoin market capitalization

Source: btctools.io, 30/06/2022



Stablecoins are on-chain digital assets designed to maintain a stable value over time **against a target asset** (e.g., a fiat currency, a commodity, a financial instrument)

Not all stablecoins are equal

Stablecoins on the market differ in some key aspects



Design



- Algorithmic
- Non algorithmic

Collateral



- Fiat-backed
- Crypto-backed
- Commodity-backed

Issuer



- Private company
- Foundation/devs
- DAO

Platform



- Platform specific
- Multichain



BORDERS

- World



CRITERIA

- Top stablecoins for market cap
- Market cap > 1 mln \$
- Extraction date 08/06/2022



SOURCE

- CoinMarketCap
- CoinGecko
- Secondary sources



56
Stablecoins



Stablecoin projects can be **designed** in a few different ways. One key distinction is **the mechanism** they use **to keep their value pegged** to the **target asset**.

Two main **design** patterns for a stablecoin:

Algorithmic



Based on a set of **smart contracts** that **govern the relationship** between the **stablecoin** and **other tokens** that back it

- **Crypto-backed Algorithmic**
- **Uncollateralized Algorithmic**
- **Partly Crypto-backed**

64%

Non Algorithmic



Collateralized by **off-chain collateral**, often held by a central entity that **issue coins redeemable** at a **1:1 ratio** for the **underlying asset**

- **Fiat-backed**
- **Commodity-backed**

36%



Stablecoins are on-chain digital assets designed to maintain a stable value over time **against a target asset** (e.g., a fiat currency, a commodity, a financial instrument)

Collateral

Many different assets can be used as collateral for backing stablecoins



Fiat-backed

- Cash equivalent
- Securities
- Bonds
- ...

29%



Commodity-backed

- Commodities
- Gold
- ...

7%



Crypto-backed

- Cryptocurrencies
- Tokens
- Other stablecoins
- ...

44%



Uncollateralized

Backed by same-network tokens

20%



Stablecoins are on-chain digital assets designed to maintain a stable value over time **against a target asset** (e.g., a fiat currency, a commodity, a financial instrument)

92%
of **total market cap** is made up
of **fiat-backed stablecoins**



Fiat-backed

- Cash equivalent
- Securities
- Bonds
- ...

29%



Commodity-backed

- Commodities
- Gold
- ...

7%



Crypto-backed

- Cryptocurrencies
- Tokens
- Other stablecoins
- ...

44%



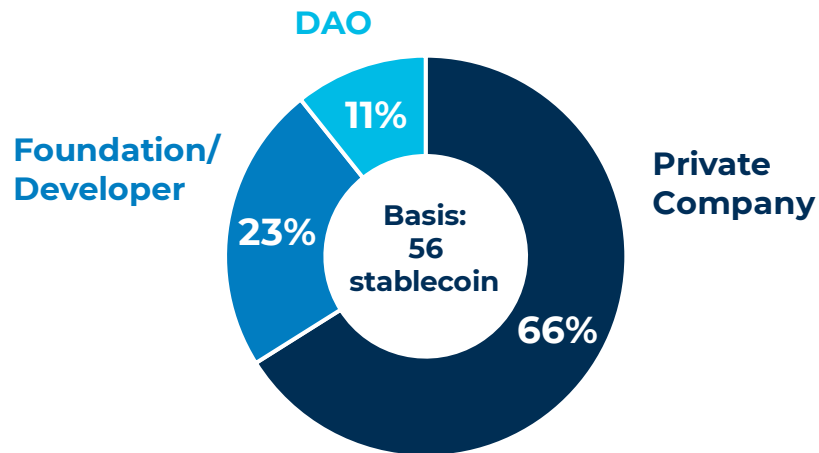
Uncollateralized

Backed by same-network tokens

20%

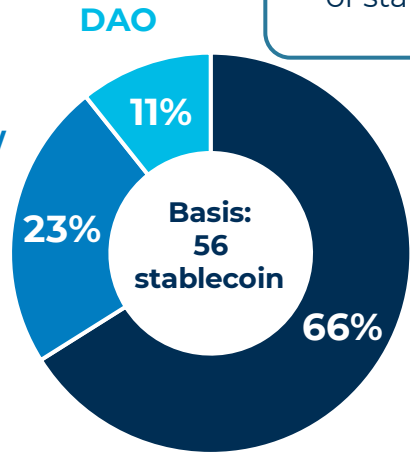


Stablecoins are primarily issued by private entities. They release "**private**" assets into the digital market, **tapping the market demand** from individuals and institutions





Stablecoins are primarily issued by private entities. They release "**private**" assets into the digital market, **tapping the market demand** from individuals and institutions



36% of stablecoins are issued by **U.S. companies**



BITFINEX



coinbase



BINANCE



Huobi



Frax



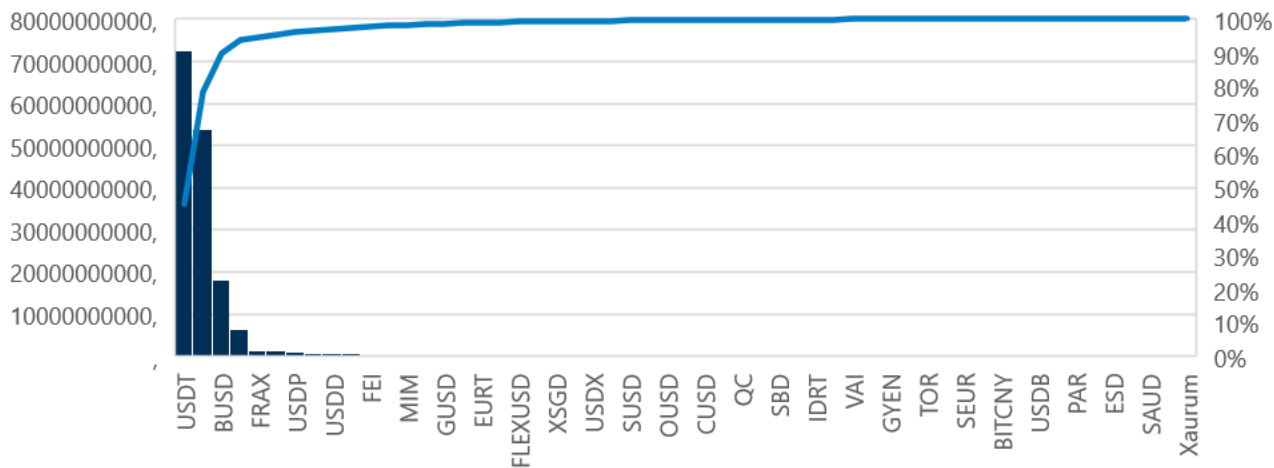
Kava

CoinFLEX.US



The **stablecoin market** is **highly concentrated**: top **5 coins** account for **95% of the total market cap**

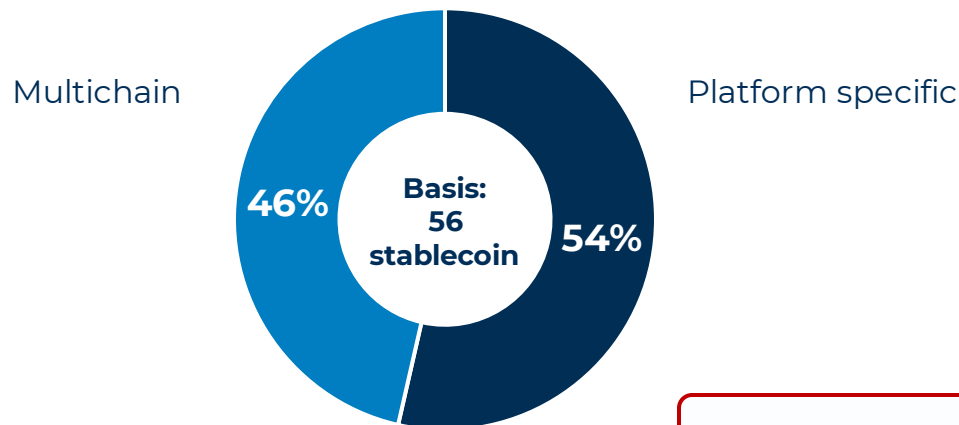
Stablecoin Market Capitalization



99%
of **total**
market cap is
made up of
stablecoins
pegged to
the **US dollar**



Almost half of the stablecoin analyzed are **multichain**. As a payment instrument that should serve many different use cases, **interoperability is a key factor**



70%
of **stablecoins** are **issued on**
Ethereum as **native platform**

	Typology	Collateral	Peg	Management of reserves	Asset custody	Market Cap	Issuer name	Issuer typology	Multi-Chain	Native Platform
 USDT	Non-algorithmic	Fiat-Backed	USD	Tether Holdings	Many Banks	72.5 Bln \$	Tether Ltd, Bitfinex	Private Company	Yes	Omni
 USDC	Non-algorithmic	Fiat-Backed	USD	Circle	Many Banks	53.9 Bln \$	Circle, Coinbase	Private Company	Yes	Ethereum
 BUSD	Non-algorithmic	Fiat-Backed	USD	Paxos	Many Banks	18 Bln \$	Paxos, Binance	Private Company	Yes	Ethereum
 DAI	Algorithmic	Crypto-Backed	USD	Maker DAO	Smart-Contract	6.6 Bln \$	Maker DAO	DAO	Yes	Ethereum
 FRAX	Algorithmic	Partly Crypto-Backed	USD	Frax Finance Team & DAO	Smart-Contract	1.5 Bln \$	Frax Finance	Private Company	Yes	Ethereum
 TUSD	Non-algorithmic	Fiat-Backed	USD	Techteryx, Ltd	Many Banks	1.3 Bln \$	Techteryx Ltd	Private Company	Yes	Ethereum
 USDP	Non-algorithmic	Fiat-Backed	USD	Paxos	Many Banks	0.9 Bln \$	Paxos	Private Company	Yes	Ethereum
 USDN	Algorithmic	Uncollateralized	USD	Neutrino Protocol Team	Smart-Contract	0.8 Bln \$	Ventuary Lab	Foundation / Developers	Yes	Waves
 USDD	Algorithmic	Crypto-Backed	USD	Tron DAO	Smart-Contract	0.7 Bln \$	Tron DAO	DAO	Yes	TRON
 PAXG	Non-algorithmic	Commodity-Backed	GOLD	Paxos	Many Banks	0.6 Bln \$	Paxos	Private Company	Yes	Ethereum



Reserve asset

CEX

Stablecoins first area of growth was within centralized exchange as a **source of liquidity**.

DeFi

The **programmability and composability of stablecoins supports decentralized finance** (DeFi). Stablecoins are a key instrument for loans and liquidity provision within decentralized protocols.



Payments

Retail

Stablecoins are used to facilitate **fast peer-to-peer and cross-border payments**. They also hold the potential for new payment innovations, such as **programmable money**. (e.g. split payment.)

Wholesale

Stablecoins facilitate **transfers of funds between firms** and allow efficient movement of internal cash across subsidiaries to manage liquidity risk and regulatory requirements.



Web3 money

Stablecoins are used to trade digital assets and serve as an **onramp from fiat currency to digital assets** recorded on blockchains. (e.g. NFTs or DvP for tokenized assets)



Reserve asset



Payments

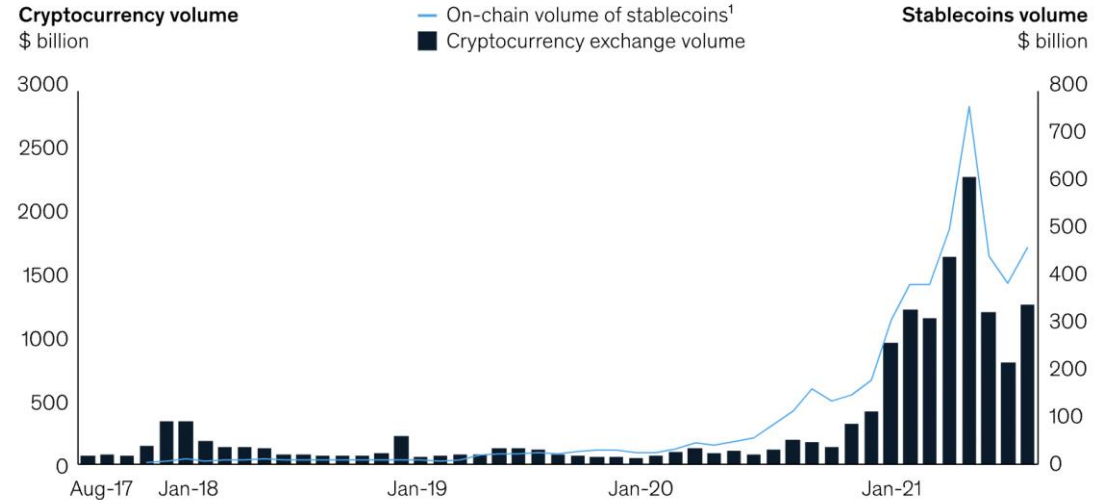


Web3 money

CEX

Stablecoins first area of growth was within centralized exchange as a **source of liquidity**.

The rise in circulation of stablecoins has closely tracked the volume of cryptocurrencies traded on exchanges over the past three years.



¹Volume of stablecoins exchanged represents all transactions recorded on the relevant blockchains. These volumes are distinct from the volume of crypto traded on exchanges, some of which may be transacted between accounts off-chain.
Source: Theblockcrypto.com

Source: McKinsey.com



What is the future of stablecoin?



How and when will **stablecoin** be **regulated**?



What are the **best design models**? Is more **transparency** needed?



What are **the scenarios of coexistence** between **stablecoin**, **crypto** and **CBDC**?

3:10 PM Stablecoins and blockchain for the exchange of value

Banks and blockchain technology: a point of view

Gloria Wan, *J.P. Morgan*

Digital money for global business

Teana Baker-Taylor, *Circle*

3:40 PM Central Bank Digital Currencies: evidence from Polimi working table

Jacopo Fracassi, *Blockchain & Distributed Ledger Observatory*

3:50 PM Central Banks' point of view

Bank of England's view on CBDC

Ben Dovey, *Bank of England*

CBDC: global trends and use cases

Ousmène Jacques Mandeng, *Accenture*

PwC Global CBDC Index

Haydn Jones, *PwC*

4:35 PM UK Government and Digital assets

Nick Davies, *International Chamber of Commerce Centre for Digital Trade and Innovation*

4:50 PM Final remarks

- **Working group: payments programmability & CBDC**
- **What's happening around the world?**
- **Stablecoins and CBDCs: features and future scenarios**

Community Osservatorio Innovative Payments

PARTNER



SPONSOR



Community Osservatorio Blockchain & Distributed Ledger

Community Osservatorio Innovative Payments

GIU
30



Highlight the fundamental aspects of programmability in the world of payments by illustrating the models that allow it to be implemented and their capabilities.

OTT
13



Describe the main design models of a possible Digital Euro currently being tested, discussing the main features about each model.

NOV
24



Highlight and collect the main considerations and indications resulting from the working group.



- **Different perception** of the **relevance of programmability** among the many players involved
- **Messy definitions** creates **confusion** among blockchain players and incumbents
- **Stablecoins** could offer some services if not provided by **CBDCs**



Secondary sources



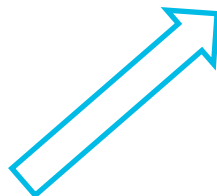
76 central banks

(>40% of total central banks worldwide)



113 initiatives

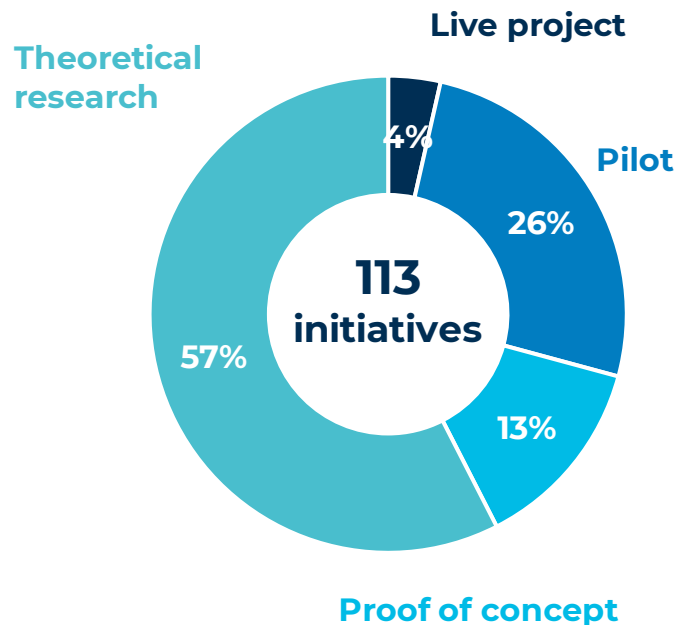
(of which **92 locals** and **21 internationals**)



21 banks have undertaken more than one local project



while **14** also participate in international projects



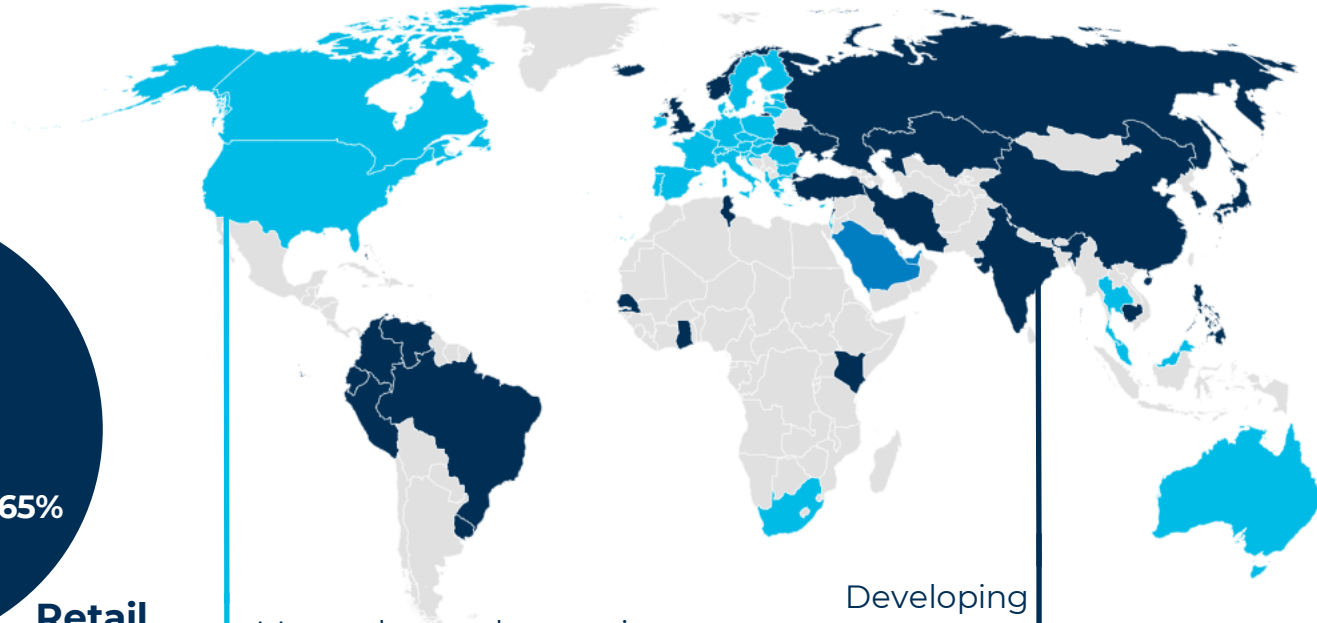
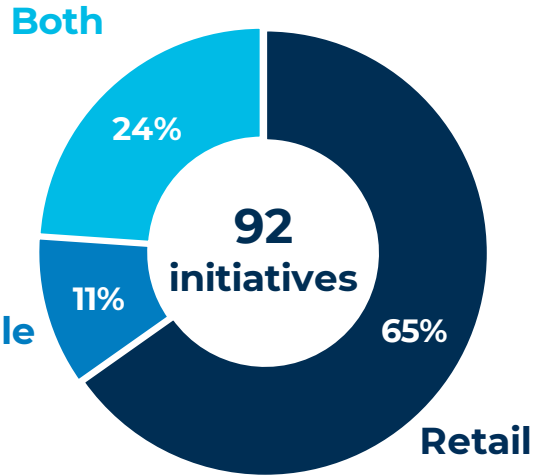
The state of advancement of projects



43% of the projects are already active with experimentation or operational (48 projects)



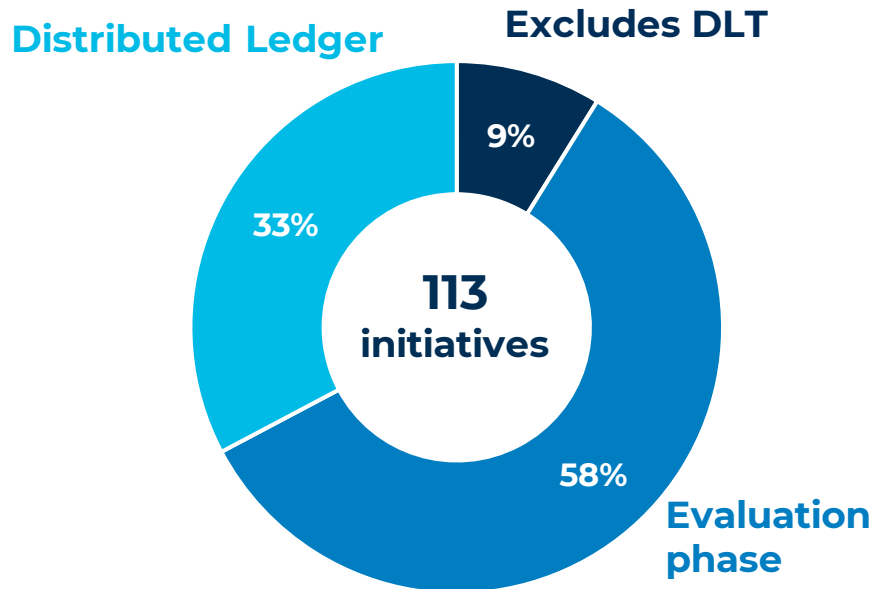
16 of the 21 international projects are in testing phase



More advanced countries such as Canada, Europe and the United States undertake both Retail and Wholesale initiatives and opt for international collaboration projects

Developing countries opt for Retail projects

Basis: 76 central banks | 92 total initiatives (not all the initiatives already have a focus)



The role of Blockchain and Distributed Ledger



91% of projects evaluate or concern this type of technology



Some of those who had initially excluded DLT have retraced their steps (e.g. e-Krona)



Assumption: CBDCs are going to be released in the market

Are CBDCs and stablecoin going to co-exist?



USA and UK are looking to adopt a measured approach with stablecoin to promote the private sector innovation while developing their CBDCs.



China's CBDC is already there however it is still in the early phases of its development.



Europe is adopting an extensive regulation (MiCAR) to regulate stablecoin and its CBDCs is expected by 2025/26 with possible limitations on it.

Programmability and **stablecoin usage** are strictly related to the design choices for a CBDC.



Stablecoins

Stablecoins are digital currencies with an exchange value pegged to a specific currency or other asset.



CBDCs

CBDCs are digital versions of fiat currencies, issued by central banks.

Programmability and **stablecoin usage** are strictly related to the design choices for a CBDC.



Stablecoins

Stablecoins are digital currencies with an exchange value pegged to a specific currency or other asset.



CBDCs

CBDCs are digital versions of fiat currencies, issued by central banks.

Why are these two extremely different instruments view as alternative solutions?



Stablecoins



CBDCs

Both these solutions can represent a **digital version of existing currencies**, giving these **specific features** that before were **only embraced by crypto-currencies**:

- Near-instantaneously **peer-to-peer** transfer of funds
- Maximum level of **programmability**
- **Delivery-versus-payment** (DvP) in tokenized financial markets



How do these instruments could compete? In which areas?



Stablecoins



Regulation



CBDCs



Design choices



How do these instruments could compete? In which areas?



Stablecoins



Regulation

Geographic area

Accessibility

Usage

Programmability



CBDCs



Design choices



How does these instruments could compete? In which areas?

Geographic area

No restrictions

Restricted

Accessibility

No restrictions

Restricted



CBDCs



Design choices

Usage

No restrictions

Restricted

Programmability

Low

Medium
(e.g. Open API
programmability
)

High
(e.g. blockchain
programmability)



How do these instruments could compete? In which areas?



Stablecoins



Geographic area

Accessibility

Usage

Programmability



CBDCs



Design choices



Stablecoins



Design choices



CBDCs

#1 Co-existence

#3 Stablecoin prominence

#2 CBDCs prominence

#1 Co-existence

Stablecoins and CBDCs co-exist in a new financial landscape providing different services.
This scenario would probably be the main one if CBDCs and stablecoins use different technological solutions.

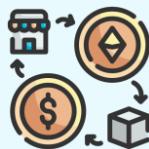
	Stablecoins	CBDCs
Geographic area	No restrictions	No restrictions
Accessibility	No restrictions	No restrictions
Usage	No restrictions	No restrictions
Programmability	High (e.g. blockchain programmability)	Low/Medium (e.g. Open API programmability)

#1 Co-existence

Stablecoins and CBDCs co-exist in a new financial landscape providing different services. This scenario would probably be the main one if CBDCs and stablecoins use different technological solutions.

Stablecoins

Stablecoins could be focused on offering access to a fiat equivalent on-chain, on-chain payments in fiat, DeFi services and liquidity provisioning.



CBDCs

CBDCs would provide direct access to central bank money in a digital form. This new solution could be also used by stablecoin providers as a collateral their reserve.



#2 CBDCs prominence

CBDCs are compatible with permissionless Blockchain technology and become used both on-chain and off-chain as the main solution to access to fiat money online.

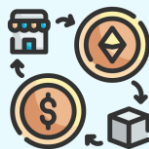
	Stablecoins	CBDCs
Geographic area	No restrictions	No restrictions
Accessibility	No restrictions	No restrictions
Usage	No restrictions	No restrictions
Programmability	High (e.g. blockchain programmability)	High (e.g. blockchain programmability)

#2 CBDCs prominence

CBDCs are in some way compatible with permissionless Blockchain technology and become used both on-chain and off-chain as the main solution to access to fiat money online.

Stablecoins

Stablecoin become a niche, some of them (the decentralized ones) would be used by a niche of users against the fiat model.



CBDCs

CBDCs provide direct access to central bank money in a digital form on-chain and off-chain. These are going to be used both in DeFi and for crypto-payments.



#2 CBDCs prominence v2

Stablecoins are heavily regulated and their usage will loss traction both on-chain and off chain.

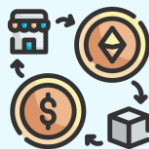
	Stablecoins	CBDCs
Geographic area	Restricted	No restrictions
Accessibility	Restricted	No restrictions
Usage	Restricted	No restrictions
Programmability	High (e.g. blockchain programmability)	Low/Medium (e.g. Open API programmability)

#2 CBDCs prominence v2

Stablecoins are heavily regulated and their usage will loss traction both on-chain and off chain.

Stablecoins

Stablecoin become a niche, some of them (the really decentralized ones without restrictions applicable) would be used by a niche of users against the fiat model.



CBDCs

CBDCs provide direct access to central bank money in a digital form. However these are going to be used mainly off-chain.



#3 Stablecoin prominence

Stablecoins capitalise their first-mover advantage and the diffusion of blockchain technology, becoming the main providers of fiat equivalents online.

	Stablecoins	CBDCs
Geographic area	No restrictions	Restricted
Accessibility	No restrictions	Restricted
Usage	No restrictions	Restricted
Programmability	High (e.g. blockchain programmability)	Low/Medium (e.g. Open API programmability)

#3 Stablecoin prominence

Stablecoins capitalise their first-mover advantage and the diffusion of blockchain technology, becoming the main providers of fiat equivalents online.

Stablecoins

Stablecoins become the main way to transact online, offering solutions on multiple chains, L1 and L2 to settle every operation happening within a Blockchain.



CBDCs

CBDCs provide direct access to central bank money in a digital form but will have restrictions among retail users, being used mainly by stablecoin providers for their reserve.





Where is Europe headed to?



Stablecoins

CBDCs

Geographic area

Restricted

Restricted

Accessibility

Restricted

Restricted

Usage

Restricted

Restricted

Programmability

High
(e.g. blockchain
programmability)

Low/Medium
(e.g. Open API
programmability)





- CBDCs and Stablecoins are **different solutions** that could solve **similar needs**
- **Design choices (CBDCs)** and **regulation (Stablecoins)** will have a huge impact on the possible diffusion of these instruments
- **Asimmetries in their design or regulation** could favour certain countries in accessing to digital services such as the ones provided through Blockchain technology